

Carl Charbel Fakhir

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Education

Georgia Institute of Technology

Bachelor of Science in Computer Science

Dec 2026

GPA: 3.84/4.00

- **Threads:** Artificial Intelligence, Information Internetworks
- **Relevant Coursework:** Data Structures & Algorithms, Computer Systems & Networks, Object-Oriented Programming, Discrete Math, AI Ethics, Linear Algebra, Probability & Statistics

Experience

NVIDIA

Incoming Software Engineer Intern

Incoming — Summer 2026

- **Selected** for the Attestation Services team to engineer secure infrastructure for GPU-accelerated AI workloads.
- **Optimizing** high-performance backend systems in **C++** for cryptographic identity and device attestation.
- **Scaling** cloud services to support remote attestation across distributed, high-density AI clusters.

PhantomPeer VPN — Miami, FL

Software Engineering Intern

Jun 2024 – Aug 2024

- **Developed** a high-performance cross-platform app using **React Native**, integrating **RESTful APIs** to handle **15,000+ monthly active connections**.
- **Built** C++ backend tooling that processed **50GB+** of network telemetry daily, enabling real-time traffic analysis.
- **Boosted** system throughput by **18%** and reduced handshake latency by **120ms** through protocol optimization.

Myki Security — Beirut, Lebanon

Cybersecurity Engineering Intern

Jun 2023 – Aug 2023

- **Secured 20,000+ daily password-vault queries** using AES-256, reducing brute-force vulnerability by **50%**.
- **Engineered C++/Python** validation tools that improved credential processing speed by **25%** during peak loads.
- **Identified and patched** 12 critical vulnerabilities including **SQLi** and **XSS** through rigorous penetration testing.

Georgia Tech — CS 3001: Computing & Society

Teaching Assistant

Aug 2025 – Present

- **Mentored 50+ students** on the intersection of AI ethics and technical systems, maintaining a **4.8/5.0** rating.
- **Streamlined** grading workflows, reducing assignment turnaround time by **15%** for faculty.

Projects

Cloud Infrastructure Cost Attribution Platform

Founder & Lead Engineer

Aug 2025 – Present

- **Architected** a cost platform for **AWS EC2, S3, and RDS** predicting expenses with **94% accuracy** via **Python/C++**.
- **Implemented MongoDB** to store **1M+ JSON** records, improving query retrieval speed by **30%** vs relational models.
- **Automated** deployment via **Docker/Kubernetes**, reducing infrastructure setup time from hours to **10 minutes**.

Distributed Compute Optimization Engine

ML Systems Engineer

Jan 2026 – Present

- **Modeled** workload prediction using CNNs, increasing GPU utilization by **22%** across distributed environments.
- **Utilized R** for statistical analysis of execution traces to identify bottleneck patterns in distributed clusters.
- **Engineered** serverless functions using **AWS Lambda** to trigger auto-scaling events based on real-time telemetry.

Leadership

LebNet & Lebanese Student Organization (LSO)

Vice President

Aug 2024 – Present

Georgia Tech

- **Leading** a community of 100+ members, coordinating professional networking events and cultural initiatives.

Skills

Programming: Java, Python, C++, C, SQL, JavaScript, R, MATLAB, Bash, JSON

Cloud & Web: AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, MongoDB, RESTful APIs, React Native, Git, CI/CD

ML & Systems: PyTorch, CNNs, KNN, Distributed Systems, Cryptography, Networking, Performance Tuning